**Ideals in the Gaussian integers**
Artin, 10.6.7

**Theorem 1.** *Every nonzero ideal in the ring of Gaussian integers contains a nonzero integer.*

*Proof.* Let $I$ be a nonzero ideal in the Gaussian integers $\mathbb{Z}[i]$. We show that there is a natural number $n$ with $0 < n$ such that $n$, viewed as an element of $\mathbb{Z}[i]$, belongs to $I$.

Since $I \neq (0)$, there exists an element $x \in I$ with $x \neq 0$. Indeed, otherwise every element of $I$ would be 0, so $I = (0)$, contradicting the assumption that $I$ is nonzero.

Write

$$x = \langle a, b \rangle = a + bi,$$

where $a, b \in \mathbb{Z}$. Since $x \neq 0$, we have

$$a \neq 0 \quad \text{or} \quad b \neq 0,$$

for if both $a = 0$ and $b = 0$, then $x = 0$.

We claim that $a^2 + b^2 > 0$. If $a \neq 0$, then $a^2 > 0$ and $b^2 \geq 0$, so $a^2 + b^2 > 0$. If instead $b \neq 0$, then $a^2 \geq 0$ and $b^2 > 0$, so again $a^2 + b^2 > 0$.

Let $n$ be the natural number given by

$$n = |a^2 + b^2|.$$

Since $a^2 + b^2 > 0$, we have $0 < n$, and casting $n$ back to $\mathbb{Z}$ gives $a^2 + b^2$.

Now, as Gaussian integers, we have

$$n = \langle a, b \rangle \cdot \langle a, -b \rangle = (a + bi)(a - bi).$$

To verify this equality, it suffices to compare real and imaginary parts. The left-hand side has real part $a^2 + b^2$ and imaginary part 0. The product on the right-hand side has real part

$$a \cdot a + (-1) \cdot b \cdot (-b) = a^2 + b^2$$

and imaginary part

$$a(-b) + ba = 0.$$

Thus the two sides are equal.

Since $\langle a, b \rangle \in I$ and $I$ is an ideal, multiplying on the right by $\langle a, -b \rangle$ keeps the element in $I$:

$$\langle a, b \rangle \cdot \langle a, -b \rangle \in I.$$

Therefore $n \in I$ as an element of $\mathbb{Z}[i]$, with $0 < n$. This gives the required nonzero integer in $I$. □